



Science — Cbse Chapter Test — 20 Marks

ANSWER KEY — FOR GRADING

Subject: Science | Class: Class 7

Total Marks: 20 | Time Allowed: 0h 40m | CBSE

General Instructions:

1. Read all questions carefully before answering.
2. All questions are compulsory unless stated otherwise.
3. Marks for each question are indicated on the right.
4. Write neatly and legibly.

Q.NO.	QUESTIONS	MARKS
Section A Attempt all 8 questions. Each carries 1 mark(s).		
1	In a kaleidoscope, why do we see different patterns each time it is turned? (A) The light source changes direction (B) The coloured pieces inside move and create new arrangements (C) The mirrors inside rotate and change angles (D) The size of the tube changes Answer: Correct Option: (B) ► Why? B. One always gets to see a different pattern every time the kaleidoscope is turned about. The coloured pieces (broken bangles or beads) move and create new arrangements through multiple reflections. [Ch 11: Light: Shadows and Reflections]	1
2	An electrician uses a screwdriver with a plastic handle and wears rubber gloves while working. Which property of plastic and rubber makes them suitable for this purpose? (A) They are good conductors of electricity (B) They are hard and durable (C) They make the tools look attractive (D) They are poor conductors of electricity and protect from electric shock Answer: Correct Option: (D) ► Why? D. Plastic and rubber are poor conductors of electricity. The plastic covering on screwdrivers and rubber gloves protect the electrician from electric shock by preventing electricity from passing through. [Ch 4: The World of Metals and Non-metals]	1



Q.NO.	QUESTIONS	MARKS
3	Which of the following metals is found in liquid state at room temperature? (A) Potassium (B) Mercury (C) Iron (D) Sodium Answer: Correct Option: (B) ► Why? B. Mercury is the metal that is found in a liquid state at room temperature. <i>[Ch 4: The World of Metals and Non-metals]</i>	1
4	A student placed an iron nail in a glass bottle with silica gel and another nail in a bottle where it was partially dipped in water with exposure to air. After 8-10 days, what observations would the student record? (A) No deposits on either nail (B) Brown deposits on the nail in the bottle with moist air; no deposits on the nail with silica gel (C) Brown deposits on both nails (D) Brown deposits on the nail with silica gel; no deposits on the nail with moist air Answer: Correct Option: (B) ► Why? B. Brown deposits (rust) develop only when iron is exposed to both air and water. The nail with silica gel (dry air only) shows no deposits, while the partially dipped nail (exposed to both) shows brown deposits. <i>[Ch 4: The World of Metals and Non-metals]</i>	1
5	Assertion (A): Mercury is a metal that exists in liquid state at room temperature. Reason (R): All metals are solid and hard at room temperature. (A) Both A and R are true, but R is NOT the correct explanation of A. (B) A is false but R is true. (C) A is true but R is false. (D) Both A and R are true, and R is the correct explanation of A. Answer: Correct Option: (C) ► Why? C. The assertion is true because mercury is explicitly stated to be a metal found in liquid state at room temperature. The reason is false because not all metals are solid and hard, as mercury is liquid and metals like sodium and potassium are soft enough to be cut with a knife. <i>[Ch 4: The World of Metals and Non-metals]</i>	1
6	Which type of object forms a darker shadow? (A) Reflective objects (B) Translucent objects (C) Opaque objects (D) Transparent objects	1



Q.NO.	QUESTIONS	MARKS
	Answer: Correct Option: (C) ► Why? C. Opaque objects form darker shadows because they block light completely, whereas translucent objects form lighter shadows. [Ch 11: Light: Shadows and Reflections]	
7	Which method of preventing rusting involves applying a protective layer of zinc metal on iron? (A) Greasing (B) Oiling (C) Galvanisation (D) Painting Answer: Correct Option: (C) ► Why? C. Galvanisation is the process of applying a protective layer of zinc metal on iron to prevent rusting. [Ch 4: The World of Metals and Non-metals]	1
8	Assertion (A): Some transparent objects can create faint shadows. Reason (R): Light passes almost completely through transparent materials. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is false but R is true. (D) A is true but R is false. Answer: Correct Option: (A) ► Why? A. Both statements are true as stated in the source. The reason correctly explains the assertion because when light passes almost completely through a transparent material, only a very small amount is blocked, resulting in a faint shadow. [Ch 11: Light: Shadows and Reflections]	1
Section B Attempt all 3 questions. Each carries 2 mark(s).		
9	What conditions are necessary for the formation of rust on iron objects? Answer: The presence of both air and water is essential for rust to form on iron objects. Iron objects kept in dry air alone or water alone do not develop brown deposits. Moist air containing both air and water is responsible for the development of rust. ► Why? This answer correctly identifies from the source material that neither dry air alone nor water alone causes rusting—both conditions together (moist air) are required. [Ch 4: The World of Metals and Non-metals]	2
10	Why is sodium metal stored in kerosene and not in water or air?	2



Q.NO.	QUESTIONS	MARKS
	Answer: Sodium reacts vigorously with oxygen and water, generating a lot of heat in the reaction. Storing sodium in kerosene prevents its exposure to moisture and air, thus protecting it from these dangerous reactions. Kerosene acts as a barrier between sodium and the environment. ► Why? This answer requires understanding why a specific storage method is used based on the reactive nature of sodium, connecting the reactivity information to the practical storage solution. [Ch 4: The World of Metals and Non-metals]	
11	What property of light is demonstrated when we cannot see a candle flame through a bent pipe? Answer: This demonstrates that light travels in a straight line. When the pipe is bent, the straight-line path of light is blocked, and we cannot see the flame. Light cannot bend around corners. ► Why? This question tests understanding of the fundamental property that light travels in straight lines, using a practical scenario described in the source material. [Ch 11: Light: Shadows and Reflections]	2
Section C Attempt 2 out of 3.		
12	How do opaque, translucent, and transparent objects differ in the type of shadows they create? Answer: Opaque objects form darker shadows because they block most of the light falling on them. Translucent objects make lighter shadows because they allow some light to pass through while still blocking a portion of it. Interestingly, even some transparent objects can create faint shadows, though they allow most light to pass through them. The difference in shadow darkness depends on how much light each type of object allows to pass through. Opaque objects block all light completely, translucent objects partially block light, and transparent objects allow almost all light to pass, resulting in faint or barely visible shadows. ► Why? The source directly states that opaque objects form darker shadows, translucent objects make lighter shadows, and some transparent objects can create faint shadows. ► Step-by-step: - Step 1: State that opaque objects form darker shadows by blocking light. - Step 2: State that translucent objects form lighter shadows by partially allowing light through. - Step 3: Mention that transparent objects can create faint shadows. [Ch 11: Light: Shadows and Reflections]	3
	OR Compare the nature of oxides formed by metals and non-metals. Explain what happens when sulfur burns in air and the resulting product is dissolved in water.	



Q.NO.	QUESTIONS	MARKS
	Answer: Metal oxides are basic in nature, whereas non-metal oxides are acidic in nature. When sulfur (a non-metal) burns in air (oxygen), sulfur dioxide gas is formed. When this sulfur dioxide gas is dissolved in water, sulfurous acid is formed, which is acidic in nature. This demonstrates that non-metal oxides produce acidic solutions. In contrast, metals react with oxygen to produce metal oxides which are basic in nature. This difference in the nature of oxides is an important distinguishing feature between metals and non-metals. ► Why? The answer correctly compares metal and non-metal oxides and describes the specific reaction of sulfur with oxygen, including the final acidic product formed. ► Step-by-step: 1. Step 1: State that metal oxides are basic while non-metal oxides are acidic. 2. Step 2: Describe the reaction: sulfur burns in air to form sulfur dioxide gas. 3. Step 3: Explain that sulfur dioxide dissolves in water to form sulfurous acid, confirming the acidic nature of non-metal oxides. [Ch 4: The World of Metals and Non-metals]	
13	When sulfur powder is heated and burned, it produces a gas. This gas is dissolved in water and tested with litmus paper. How does the solution formed from sulfur oxide differ from the solution formed from magnesium oxide, and what does this tell us about the difference between metals and non-metals? Answer: When sulfur burns, it produces a gas that dissolves in water to form a solution. When tested with litmus paper, this solution behaves differently from magnesium oxide solution. While magnesium oxide solution turns red litmus to blue (indicating basic nature), the solution from sulfur oxide would show the opposite effect - turning blue litmus to red, indicating an acidic nature. This demonstrates a fundamental difference between metals and non-metals: oxides of metals are basic in nature, whereas oxides of non-metals are acidic in nature. Magnesium is a metal, so its oxide is basic. Sulfur is a non-metal, so its oxide is acidic. This difference in the chemical nature of oxides helps distinguish metals from non-metals.	3



Q.NO.	QUESTIONS	MARKS
	► Why? The answer compares the acidic nature of sulfur oxide with the basic nature of magnesium oxide and relates this to the fundamental difference between metals and non-metals. ► Step-by-step: 1. Step 1: Recall that magnesium oxide solution is basic and turns red litmus blue. 2. Step 2: Understand that sulfur is a non-metal and its oxide solution would be acidic, turning blue litmus red. 3. Step 3: Conclude that metal oxides are basic while non-metal oxides are acidic – a key difference between metals and non-metals. [Ch 4: The World of Metals and Non-metals]	

